Grade: Check Plus

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Assignment #1

The probability distribution of the amount drawn from an ATM at a convenience store is tabulated below. In this table X = the amount withdrawn by a randomly selected customer and f(x) = P[X = x]. A customer can only withdraw the amounts indicated in the first column of the table below.

x	f(x)
10	0.10
20	0.15
40	0.30
60	0.15
80	0.25
100	0.05

a. **What is the probability that a randomly selected customer withdraws at least $60 from the ATM?**

✓ Qa

Probability of $60 or greater is the sum of the probability of $60 + $80 + $100 = 0.15 + 0.25 + 0.05 = 0.45 or 45% chance of a customer withdrawing $60 or more

b. **Show by calculation that the expected value withdrawn from the ATM is $50 and that the standard deviation of the amount withdrawn from this ATM is $26.27.**

The expected value to be withdrawn is equal to E(X) = ∑x f(x) = 10(0.10) + 20(0.15) + 40(0.30) + 60(0.15) + 80(0.25) + 100(0.05) = 50

✓ Qb

Variance is equal to the sum of x^2 * f (X) minus the square of the expected value = (10^2(0.10) + 20^2(0.15) + 40^2(0.30) + 60^2(0.15) + 80^2(0.25) + 100^2(0.05))-50^2 = 3190 – 2500 = 690

Standard Deviation is equal to the Square Root of the Variance = $\sqrt{690}$ = $26.27

c. **The manager of this convenience store tells you that an expected value of $50 is not correct because it is impossible to withdraw $50 from the ATM. How would you answer the manager?**

✓ Qc

Even though it is not possible to draw $50 cash from the ATM, over a large enough sample size, the expected value would work out to $50, due to the weighted probabilities of a given user taking out a given amount of money. The expected value is the average amount across all transactions.

d. **What is the probability that the amount withdrawn from the ATM will fall within one standard deviation of the mean?**

Expected Value = 50, StDev = 26.27. 50±26.27 = 23.73 and 76.27. Values in this range are $40 and $60. The probability of either $40 or $60 is 30% + 15% = 45% chance that a withdrawal will fall within one SD of the mean ✓ Qd

e. **What is the probability that the amount withdrawn from the ATM will fall within two standard deviations of the mean?**

Expected Value = 50, StDev = 26.27. 50±2(26.27) = -2.54 and 102.54. Values in this range are $10, $20, $40, $60, $80, and $100. As this is the entire range of possible withdrawals, the probability that a given withdrawal is within 2SD of the mean is 100% ✓ Qe